

HOSPITAL MANAGEMENT SYSTEM

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1.Introduction

○ Purpose:

The purpose of this document is to describe the functional and non-functional requirements of the Hospital Management System (HMS). This SRS serves as a reference for developers, testers, instructors, and stakeholders to understand system behaviour, constraints, and expectations. It acts as a contractual document between the client and the development team.

○ Document Conventions:

- HMS – Hospital Management System
- Admin – Hospital administrator
- User – Doctor, receptionist, or staff
- All requirements are stated using “shall” to indicate mandatory behaviour.
- Database terms follow standard SQL/MySQL conventions.

○ Intended Audience and Reading Suggestions:

a) This document is intended for:

- Project evaluators and faculty
- Software developers
- Database designers
- Testers and maintainers

b) Reading Suggestions:

- Read **Overall Description** for system understanding
- Refer **System Features** for functionality
- Use **Nonfunctional Requirements** for quality constraints

○ Project Scope:

The Hospital Management System is a web-based application designed to automate hospital operations such as:

- Patient registration
- Doctor management
- Appointment scheduling
- Billing and payment records • Medical record maintenance

The system improves efficiency, reduces paperwork, and ensures secure data handling.

○ References:

- IEEE Std 830-1998 – Software Requirements Specification
- MySQL Documentation
- PHP Official Documentation
- Software Engineering Textbook (Pressman)

2.Overall Description

○ Product Perspective:

The HMS is a standalone web application that interacts with a centralized MySQL database. It replaces manual hospital record management and integrates patient, doctor, and billing modules into a single system.

○ Product Features:

- Patient registration and record management
- Doctor and staff management
- Appointment scheduling
- Billing and payment tracking
- Secure login and role-based access
- Report generation

○ User Classes and Characteristics:

User Type	Characteristics
Admin	Manages system, users, and reports
Doctor	Views patient details and appointments

Receptionist	Registers patients and schedules appointments
Billing Staff	Handles billing and payments

○ **Operating Environment:**

- Server: Apache / XAMPP
- Backend: PHP
- Database: MySQL
- Frontend: HTML, CSS, JavaScript
- Browser: Chrome, Firefox, Edge
- OS: Windows / Linux

○ **Design and Implementation Constraints:**

- Must use PHP and MySQL
- Web-based architecture
- Limited hardware resources
- Must follow relational database design principles

○ **User Documentation:**

- User manual
- Admin guide
- Installation guide
- Online help pages

○ Assumptions and Dependencies:

- Internet or local server availability
- Authorized users only
- Database server is operational
- Users provide correct input data

3. System Features

○ System Feature 1: Patient Management:

‡ Description:

Allows registration, updating, and viewing of patient records.

‡ Functional Requirements:

- The system shall allow staff to register new patients.
- The system shall store patient personal and medical details.
- The system shall allow updating and deletion of patient records.
- The system shall retrieve patient history when required.

○ System Feature 2: Appointment & Billing Management:

‡ **Description:**

Handles appointment scheduling and billing operations.

‡ **Functional Requirements:**

- The system shall allow appointment booking with doctors.
- The system shall avoid appointment conflicts.
- The system shall generate billing details.
- The system shall maintain payment records.

4. External Interface Requirements

○ User Interfaces:

- Web-based GUI
- Forms for patient, doctor, and billing details
- Dashboards for admin and staff
- Simple and responsive design

○ Hardware Interfaces:

- Server system
- Client computers
- Keyboard, mouse, and display devices

○ **Software Interfaces:**

- MySQL database
- PHP server
- Web browser

○ **Communications Interfaces:**

- HTTP/HTTPS protocol
- Localhost or LAN network
- TCP/IP for database communication

5. Other Nonfunctional Requirements

○ **Performance Requirements:**

- System shall support multiple users simultaneously.
- Response time shall be less than 3 seconds for normal operations.

○ **Safety Requirements:**

- Data backup mechanisms shall be implemented.
- System shall prevent accidental data loss.

○ Security Requirements:

- Secure login with authentication
- Role-based access control
- Password encryption
- Protection against unauthorized access

○ Software Quality Attributes:

- **Reliability:** System works consistently
- **Usability:** Easy to use interface
- **Maintainability:** Easy to update and modify
- **Scalability:** Supports future expansion

6. Other Requirements:

- Audit logs for major operations
- Data validation on all input forms
- Error handling and user-friendly messages

7. Appendix A: Glossary

Term	Description
HMS	Hospital Management System

Admin	System Administrator
DBMS	Database Management System
UI	User Interface

8. Appendix B: Analysis Models

- Entity Relationship (ER) Diagram
- Data Flow Diagrams (Level 0 & Level 1)
- Use Case Diagrams
- Relational Schema

9. Appendix C: Issues List

- Integration with external lab systems (future scope)
- Performance optimization for large databases
- Mobile application support

10.Result

Code part :

Database Connection (PHP)

```
<?php  
$con = mysqli_connect("localhost","root","","hms");  
  
if(!$con){  
    die("Connection Failed");  
}  
?>
```

Patient Login Code

```
if(isset($_POST['login']))  
{  
    $email = $_POST['email'];  
    $password = md5($_POST['password']);  
  
$query = mysqli_query($con,"select ID from tblpatient  
where Email='$email' && Password='$password'");  
  
$ret = mysqli_fetch_array($query);  
  
if($ret>0){  
    $_SESSION['pid']=$ret['ID'];  
    header('location:dashboard.php');  
}
```

Appointment Booking Code

```
if(isset($_POST['submit']))  
{  
$doctor = $_POST['doctor'];  
$date = $_POST['date'];  
$time = $_POST['time'];  
  
$query = mysqli_query($con,"insert into appointment  
(doctorName,appointmentDate,appointmentTime)  
values('$doctor','$date','$time')");  
}
```

Add Doctor Code (Admin Module)

```
if(isset($_POST['submit']))  
{  
$name=$_POST['doctorname'];  
$spec=$_POST['specialization'];  
  
$query=mysqli_query($con,"insert into doctors  
(doctorName,specialization)  
values('$name','$spec')");  
}
```

AI Symptom Checker Logic

```
if($symptom=="chest pain")  
echo "Consult Cardiologist";
```

```
elseif($symptom=="skin allergy")  
echo "Consult Dermatologist";
```

```
elseif($symptom=="headache")  
echo "Consult Neurologist";
```

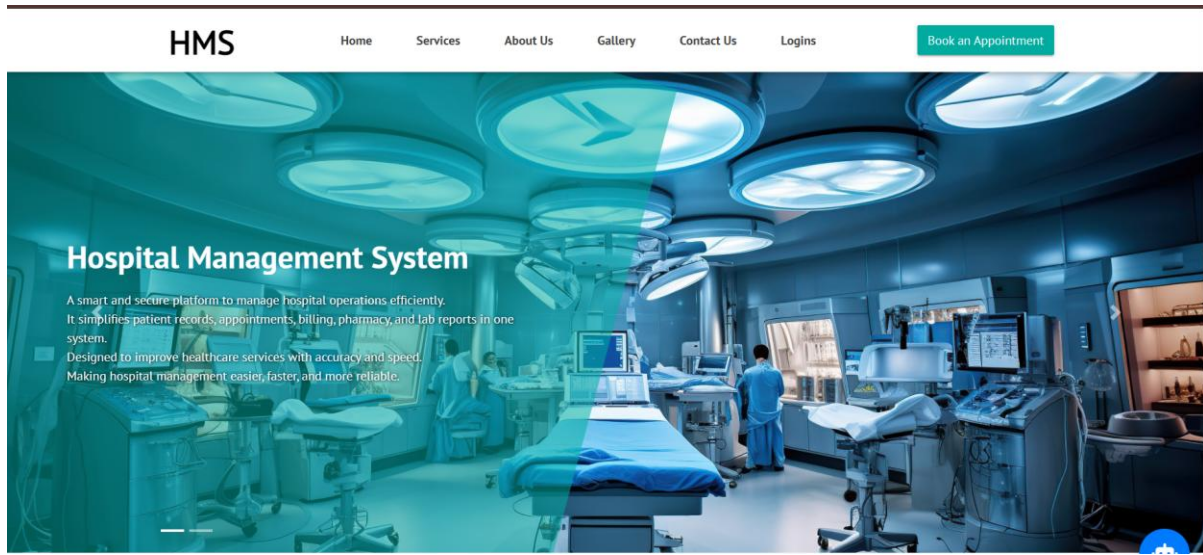
Display Patient Records

```
$query=mysqli_query($con,"select * from tblpatient");
```

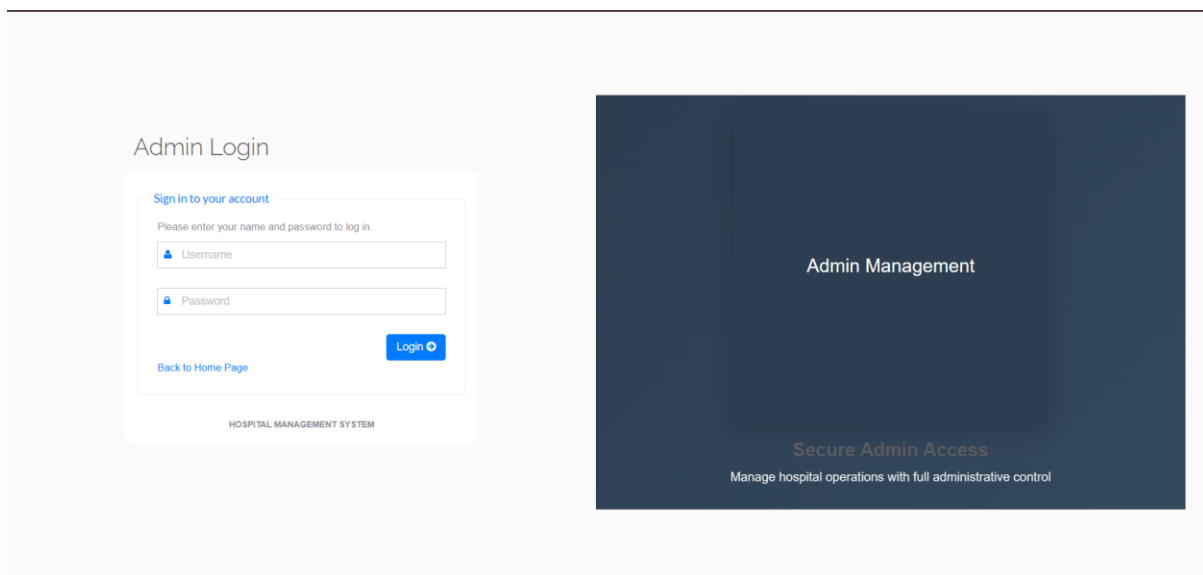
```
while($row=mysqli_fetch_array($query))  
{  
echo $row['PatientName'];  
echo $row['Email'];  
}
```

Output :

Home page :



Admin Portals :



Patient Portal :

HMS | Patient Login

Sign in to your account

Please enter your name and password to log in.

Email

Password

[Forgot Password ?](#)

Login

Don't have an account yet? [Create an account](#)

HOSPITAL MANAGEMENT SYSTEM

Your Health Matters

Access your medical records and book appointments easily

Doctors Portal :

HMS | Doctor Login

Sign in to your account

Please enter your name and password to log in.

Email

Password

[Forgot Password ?](#)

Login

HOSPITAL MANAGEMENT SYSTEM

Medical Expertise

Professional Care

Manage patient care and medical records with ease

Insurance Checker :

The screenshot displays the HMS | Health Insurance Portal. On the left, there is a form titled "Insurance Status Check" with the instruction "Please enter your Aadhar number to check your insurance status." Below this is a text input field labeled "Aadhar Number (12 digits)" and a blue "Check Status" button. The form is set against a light gray background with the text "HEALTH INSURANCE PORTAL" at the bottom. To the right, there is a large blue and purple banner with the text "Health Insurance" in white, "Secure Your Health" in green, and "Check your insurance coverage instantly with your Aadhar number" in white at the bottom.

AI chatbot :

The screenshot shows the AI Symptom Assistant chatbot interface. The chatbot has a white header with the title "AI Symptom Assistant" and a close button (X). Below the header, there is a text input field with the placeholder text "Describe your symptoms (type or use microphone):" and a microphone icon. The input field contains the text "e.g. chest pain and shortness of breath". To the right of the input field is a green "Ask" button. Below the input field, there is a message that says "Speak or submit symptoms to get recommended doctors." The chatbot is displayed over a background that includes a "Doctors Login" button and a "Admin Login" button.